❖ DETAILED LESSON PLAN ❖

Factors & Multiples

What Are Factors & Multiples?

Subject	Mathematics — Number Concepts
Session	Session 1 of 1
Duration	30 Minutes
Level	Beginner
Board	CBSE / ICSE / State Boards (India)
Medium	English (child-friendly)
Characters	Arjun & Priya (student guides throughout)

LEARNING OBJECTIVES

LO 1	Understand what a FACTOR is — a number that divides another exactly with no remainder.
LO 2	Understand what a MULTIPLE is — a number obtained by multiplying by 1, 2, 3…
LO 3	Find all factors of numbers up to 20 using equal-grouping.
LO 4	Identify the first five multiples of numbers 1 through 10.
LO 5	Distinguish clearly between a factor and a multiple using real examples.
LO 6	Recognise that every number has at least two factors: 1 and itself.
LO 7	Understand that multiples go on forever (infinitely).
LO 8	Use skip counting to find multiples.

SECTION 1 — WARM-UP ACTIVITY (0–3 min)

Arjun asks: "Amma has 12 laddoos.
Can we share them equally with friends? How many ways can we make equal groups?"

Skip Counting Warm-up:

Ask students to count aloud in 2s: **2, 4, 6, 8, 10, 12**. Then in 3s: **3, 6, 9, 12**. Demonstrate equal groups physically using counters, fingers, or drawing circles on the board.

Discussion prompt (30 seconds):

"What are all the ways we can share 12 laddoos equally — no leftover!" Accept all responses. This naturally leads into the definition of factors.

SECTION 2 — WHAT IS A FACTOR? (3–8 min)

Definition (say aloud, write on board):

"A factor is a number that divides another number exactly — with NO leftover!"

Four Golden Rules of Factors:

- **Rule 1:** Think of factors as "equal groups" of something.
- **Rule 2:** Factors are always smaller than or equal to the number.
- **Rule 3:** 1 is a factor of EVERY number.
- **Rule 4:** Every number is a factor of itself.

Key Formula:

If A × B = C ➜ then A and B are BOTH factors of C

Classroom Examples with Indian Context:

■ **Laddoos**	12 laddoos in equal rows: 1×12, 2×6, 3×4, 4×3 Factors of 12 = 1, 2, 3, 4, 6, 12

■ **Cricket**	10 players split into equal teams: 2 teams of 5 ✓ 3 teams? ✗ (leftover!) 2 and 5 are factors of 10; 3 is NOT
■ **Diyas**	16 diyas arranged in rows: 1×16, 2×8, 4×4 Factors of 16 = 1, 2, 4, 8, 16
■ **Coconuts**	9 coconuts in 3 baskets of 3 each: 3×3 = 9 Factors of 9 = 1, 3, 9

SECTION 3 — HOW TO FIND FACTORS (8–13 min)

Step-by-Step Method (the "Pair Method"):

Step 1	Start with 1. Write: 1 × ? = your number.
Step 2	Try 2. Does it divide evenly? If yes, write both.
Step 3	Try 3, 4, 5… Continue until you reach the middle.
Step 4	Stop when the factor pairs start repeating.

Worked Example — Factors of 12:

1 × 12 = 12	✓ Both 1 and 12 are factors
2 × 6 = 12	✓ Both 2 and 6 are factors
3 × 4 = 12	✓ Both 3 and 4 are factors
4 × 3 = 12	■ Same pair — STOP here!

Answer: Factors of 12 = 1, 2, 3, 4, 6, 12 (six factors total)

CLASSROOM PRACTICE — SET A (Finding Factors)

Find all factors of the following numbers using the Pair Method. Write your working in the space below each question.

Factors of 8	= ______, ______, ______, ______	*Ans: 1, 2, 4, 8*
Factors of 15	= ______, ______, ______, ______	*Ans: 1, 3, 5, 15*
Factors of 20	= ______, ______, ______, ______, ______, ______	*Ans: 1, 2, 4, 5, 10, 20*
Factors of 18	= ______, ______, ______, ______, ______, ______	*Ans: 1, 2, 3, 6, 9, 18*

SECTION 4 — WHAT IS A MULTIPLE? (13–18 min)

**Priya says: "Multiples are like stepping stones!
Each step adds the SAME number. And they go on forever!"**

Definition:

"A multiple is what you get when you MULTIPLY a number by 1, 2, 3, 4…"

Four Key Facts About Multiples:

- Multiples are always equal to or BIGGER than the original number.
- Found by multiplying (NOT dividing).
- Every number has INFINITE multiples — they never stop!
- The first multiple of any number is the number itself (×1).

Indian Examples — Multiples in Everyday Life:

■ **Rotis per meal**	Multiples of 3: **3, 6, 9, 12, 15...** *If 3 rotis are made per meal, how many after 2, 3, 4, 5 meals?*
■ **Rickshaw wheels**	Multiples of 3: **3, 6, 9, 12, 15...** *3 wheels per auto-rickshaw: 1 rickshaw=3, 2=6, 3=9 wheels...*
■ **■5 coins**	Multiples of 5: **5, 10, 15, 20, 25...** *Count ■5 coins: ■5, ■10, ■15, ■20, ■25...*
■ **Biscuit packets**	Multiples of 6: **6, 12, 18, 24, 30...** *Each packet has 6 biscuits: 1 pkt=6, 2=12, 3=18...*

CLASSROOM PRACTICE — SET B (Finding Multiples)

Write the first 5 multiples of each number by multiplying by 1, 2, 3, 4, 5:

Multiples of 4:	______ , ______ , ______ , ______ , ______	*Ans: 4, 8, 12, 16, 20*

Multiples of 6:	______ , ______ , ______ , ______ , ______	*Ans: 6, 12, 18, 24, 30*
Multiples of 7:	______ , ______ , ______ , ______ , ______	*Ans: 7, 14, 21, 28, 35*
Multiples of 9:	______ , ______ , ______ , ______ , ______	*Ans: 9, 18, 27, 36, 45*

SECTION 5 — FACTORS vs MULTIPLES (18–22 min)

This is the most important section — helping students tell the two apart.

FACTORS

- Smaller or equal to the number
- Found by DIVIDING
- FINITE (few of them)
- Example: Factors of 6 = 1, 2, 3, 6
- Start from 1

MULTIPLES

- Equal or LARGER than the number
- Found by MULTIPLYING
- INFINITE (go on forever!)
- Example: Multiples of 6 = 6,12,18,24...
- Start from the number itself

Using the SAME number — 6:

	Factors of 6	Multiples of 6
Numbers	1, 2, 3, 6	6, 12, 18, 24, 30, 36…
How many?	Only 4 (finite)	Endless (infinite)
Size	All ≤ 6	All ≥ 6
How found?	Divide 6 by 1, 2, 3…	Multiply 6 × 1, 2, 3…

MEMORY TRICK — "Factors FIT, Multiples MARCH!"

FACTORS FIT inside!	MULTIPLES MARCH on forever!
"Factors are FEW and SMALL — they FIT inside the number!"	"Multiples are MANY and BIG — they MARCH on without stopping!"

SECTION 6 — TRUE OR FALSE QUIZ (22–26 min)

Read each statement aloud. Students hold up a green card (True) or red card (False). Then discuss WHY.

1.	Factors are always smaller than or equal to the number.	**TRUE ✓**	*1 and the number itself are always factors; both are ≤ the number.*
2.	Multiples stop after 10.	**FALSE ✗**	*Multiples go on forever! 10 is just one multiple — they continue 11×, 12×…*
3.	1 is a factor of every number.	**TRUE ✓**	*1 × any number = that number, so 1 always divides perfectly.*
4.	Multiples are found by dividing.	**FALSE ✗**	*Multiples are found by MULTIPLYING. Factors are found by dividing.*
5.	12 is a multiple of 4.	**TRUE ✓**	*4 × 3 = 12, so yes — 12 is the 3rd multiple of 4.*
6.	5 is a factor of 14.	**FALSE ✗**	*14 ÷ 5 = 2 remainder 4. There is a leftover, so 5 is NOT a factor of 14.*

SECTION 7 — MULTIPLE CHOICE ASSESSMENT (26–30 min)

Name: ______________________ **Date:** ____________ **Score:** _____ / 5

Q1. Which of the following is a factor of 18?

A) 7 B) 4

C) 6 ✓ D) 8

Answer: C — 18 ÷ 6 = 3 exactly, no remainder.

Q2. What is the 4th multiple of 5?

A) 15 B) 25

C) 10 D) 20 ✓

Answer: D — 5 × 4 = 20.

Q3. Priya has 24 diyas. Which is NOT a factor of 24?

A) 8 B) 6

C) 7 ✓ D) 4

Answer: C — 24 ÷ 7 = 3 remainder 3, so 7 is NOT a factor.

Q4. Which statement about multiples is TRUE?

A) Multiples stop at 100
B) Multiples are found by dividing
C) Multiples go on forever ✓
D) Multiples are always less than the number

Answer: C — Multiples never stop; they are infinite.

Q5. How many factors does 12 have?

A) 3
B) 4
C) 5
D) 6 ✓

Answer: D — Factors of 12: 1, 2, 3, 4, 6, 12 — six factors.

LET'S RECAP TODAY'S LEARNING!

Factors	✓ divide numbers exactly — no remainder. They are FEW and SMALL.
Multiples	✓ are found by multiplying and go on FOREVER. They are MANY and BIG.
Rule 1	✓ 1 is a factor of EVERY number.
Rule 2	✓ Every number is a factor of itself.
Rule 3	✓ Multiples of a number are infinite — they never stop!
Memory	✓ Factors FIT inside. Multiples MARCH on forever!

TEACHER NOTES & FACILITATION TIPS

- **Warm-up:** Use physical counters or draw circles on the board when sharing laddoos.
- **Factors Activity:** Have students work in pairs — one tries to divide, one records.
- **Multiples Activity:** March around the classroom counting in steps! Physical movement helps.
- **Misconception 1:** Students often confuse factors with multiples. Repeat the memory trick often.
- **Misconception 2:** Some think 1 is not a factor — always reinforce: 1 × N = N, so 1 IS a factor.
- **Misconception 3:** Students think factors stop at a specific number — clarify with examples.
- **Extension:** Ask: "What is the largest factor of any number? The number itself!"
- **Visual Aid:** Use array diagrams: 12 objects arranged in rows to show all factor pairs visually.
- **Assessment:** The MCQ can be done as an exit card — 5 questions, 5 minutes.

COMMON MISCONCEPTIONS TO ADDRESS

Misconception	Correction	How to Respond
"Factors go on forever"	Factors are FINITE	"Try listing factors of 6 — there are only 4!"
"Multiples are found by dividing"	Multiples need MULTIPLICATION	"Watch: 3×1=3, 3×2=6, 3×3=9 — that's multiplying!"
"1 is not a factor"	1 IS a factor of every number	"1 × 12 = 12, so 1 divides 12 perfectly!"
"Every number is a multiple of 1"	Correct — reinforce this!	"Yes! Because N×1=N, every number is a multiple of 1."

HOMEWORK / TAKE-HOME ACTIVITY

- Find all factors of 16 and 25. Write your working using the Pair Method.
- Write the first 8 multiples of 3 and 7.

- ♦ BONUS: Ask a family member to pick any number under 20. Find all its factors!
- ♦ INDIAN CONTEXT: Count how many ways you can arrange 24 flowers in equal rows for Diwali decoration.

WHAT'S NEXT — PREVIEW OF SESSION 2

- Finding factors and multiples of BIGGER numbers (up to 100).
- Introduction to Common Factors — what two numbers share.
- Introduction to Common Multiples — where two number-lists overlap.
- Fun games and challenges with Arjun and Priya!

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